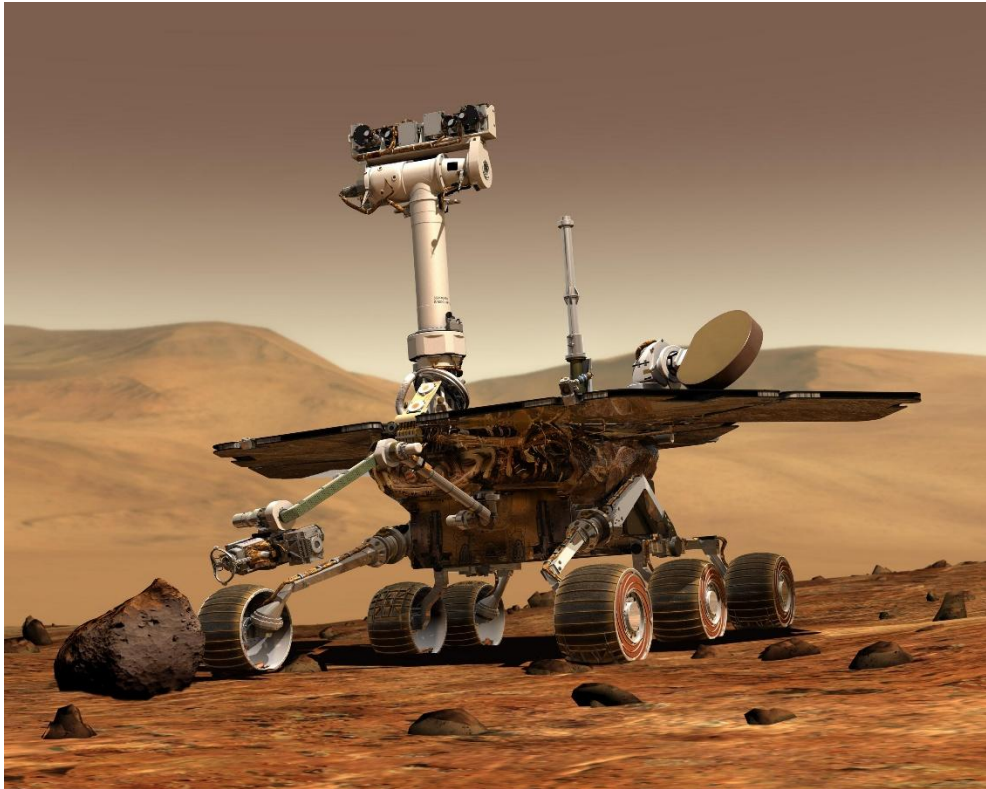


SoilSpirit

Autonomous No-dig Farming Rover



Appendix A

Creator:

Damon Rocha, Robotics and Embedded Systems

Last Updated: 5/17/2025

Technical Field

This project focuses on the following technical fields: robotics, embedded systems, autonomous systems, and agriculture.

Background Information

No dig farming is a method of growing food where the soil is undisturbed. Instead of digging into the ground, compost is spread evenly on top of the soil and the seeds are directly put into this layer of material. It leads to healthy plants and soil, reduces weeds, and allows for better water conditions for the plants (Dowding, 2024). SoilSpirit plans to reduce the physical strain on small farms and gardeners through the automation of compost and seed spreading. While also having a small enough footprint that it does not compact the soil or cause unwanted tilling/digging of the compost. It takes inspiration from simple autonomous vehicles for obstacle avoidance, Mars rovers (Spirit, Curiosity, Perseverance) for its rocker no bogie suspension system, and a new cycloidal gear system for maximizing the pulling power of the rover at a small scale (RoTechnic, 2022).

Prior Art (legal term)

The design of the product was heavily influenced by the Mars rovers and the Instructables tutorial in AppendixC. This was much truer of the initial prototype, for the final system I completely rebuilt it from scratch and now only the trailer uses parts from this resource. The suspension system of this rover comes from a simplification of the Rocker Bogie Mechanism popularized by the Mars rovers Curiosity and Perseverance. It removes the bogie arm and just uses rockers on either side of the rover's body. Finally, a youtube video, that walked through a single tooth cycloidal gear system was used to start the build of the final drive system (RoTechnic, 2022). I took this and created a drive system around it to work with wheels found at this resource (AurthurGuy, 2018). As for similar products, the closest to my idea is an autonomous spreader built by Raven Industries. It is an industrial vehicle that is autonomous. It is great for automation for large scale farms but would not benefit gardeners, homesteaders, and small-scale farms wanting to use no dig gardening. Another similar product that is of the same form factor/user is FarmBot. A CNC based garden bed that automates the process of growing food entirely. This company will be a direct competitor with the main difference being our modes of distribution. Their machine is directly attached to a garden bed whereas mine can drive around and be used for different types of gardens, farms, and homesteads.

Project Description

The aim of the project is to develop an autonomous rover capable of spreading fertilizer and seeds over a garden while partially avoiding obstacles. It should be able to:

- Drive on uneven surfaces

- Pull ~1lb of fertilizer
- Evenly spread materials
- Autonomously stop rover when detecting obstacles using ultrasonic sensor
- Try to maneuver around obstacles using TOF sensor
- Move ~ 2mph while not hauling spreader

It will consist of the following components:

1. Chassis to house electronics
2. 2 L298N motor driver
3. 4 DC Motors
4. ESP32 ESP-S Dev Kit
5. 3 3.7V Lip rechargeable battery
6. Power bus with fuses
7. VL53L5CX TOF sensor
8. Hopper trailer with spring-based trap door to spread fertilizer/seeds

To demonstrate the system's capabilities, I plan on first running this on a mock test area in my garage. Then I want to move on and try it with freshly cut grass in my backyard and get it to spread fertilizer until I have issued a stop command.

Innovation Claim

SoilSpirit will be the first adaptation of Mars rover rocker-bogie suspension for agriculture and will make automation accessible to more people. On top of that it promotes a sustainable agricultural practice in making no-dig farming easier. Lastly, it will provide users with a

Usage Scenario

The primary demographic of this product encompasses gardeners, homesteaders, and small farm owners who are interested in no-dig farming or already practice it. However, this system could easily be extended including applications across industry. In aerospace, this may very well become integrated with future iterations of rovers and space vehicles. Technology inspired by SoilSpirit could possibly be used for farming on other planets, or as small-scale versions of the Mars rovers that are more accessible for smaller companies. SoilSpirit could also extend into disaster recovery and reconnaissance. A scaled-up version of this rover could be used for search and rescue, area mapping, and debris clearing. Additionally, this project would be valued by the education and start-up industry due to it being a starting point for various projects.

Evaluation Criteria

The following questions will identify the successful completion of the project:

1. Core System Functionality:
 - a. Can the rover smoothly drive without user intervention?
 - b. Can the rover move at least 2MPH while not hauling the hopper?
 - c. Does the hopper start/stop spreading based on movement of the rover?
 - d. Does the rover successfully detect obstacles and either avoid or stop before hitting them?
 - e. Can the rover drive on slightly uneven surfaces?
2. Motor control and turning:
 - a. Can the motors drive the rover forward and backwards?
 - b. Can the motors cause the rover to turn left/right?
 - c. Is there enough charge to run the motors to allow for spreading to finish?
3. Object Detection:
 - a. Do the emergency stop TOF detect obstacles?
 - b. Does a detection lead to the rover stopping?
 - c. Does TOF sensor detect objects in different regions?
 - d. Can the rover go around the object?

Objectives and Tasks Associated with the Project

High-level Objectives

1. Development of Motor control and rocker-bogie suspension system
 - a. Create 2 rocker no bogie suspension systems
 - b. Connect to microcontroller to test control
 - c. Test DC motors, servos, and full suspensions systems individually then as a system
 - d. Setup interface to drive motors based on shared commands
2. Obstacle detection
 - a. Test TOF and determine correct ranges and grid
 - b. Test detection of different objects, conditions, and distances
 - c. Test TOF object region detection and partial obstacle avoidance
3. Rover control and drive system
 - a. Integrate object detection into motor control system
 - i. ON/OFF/start/stop/left/right commands from motor control system interface

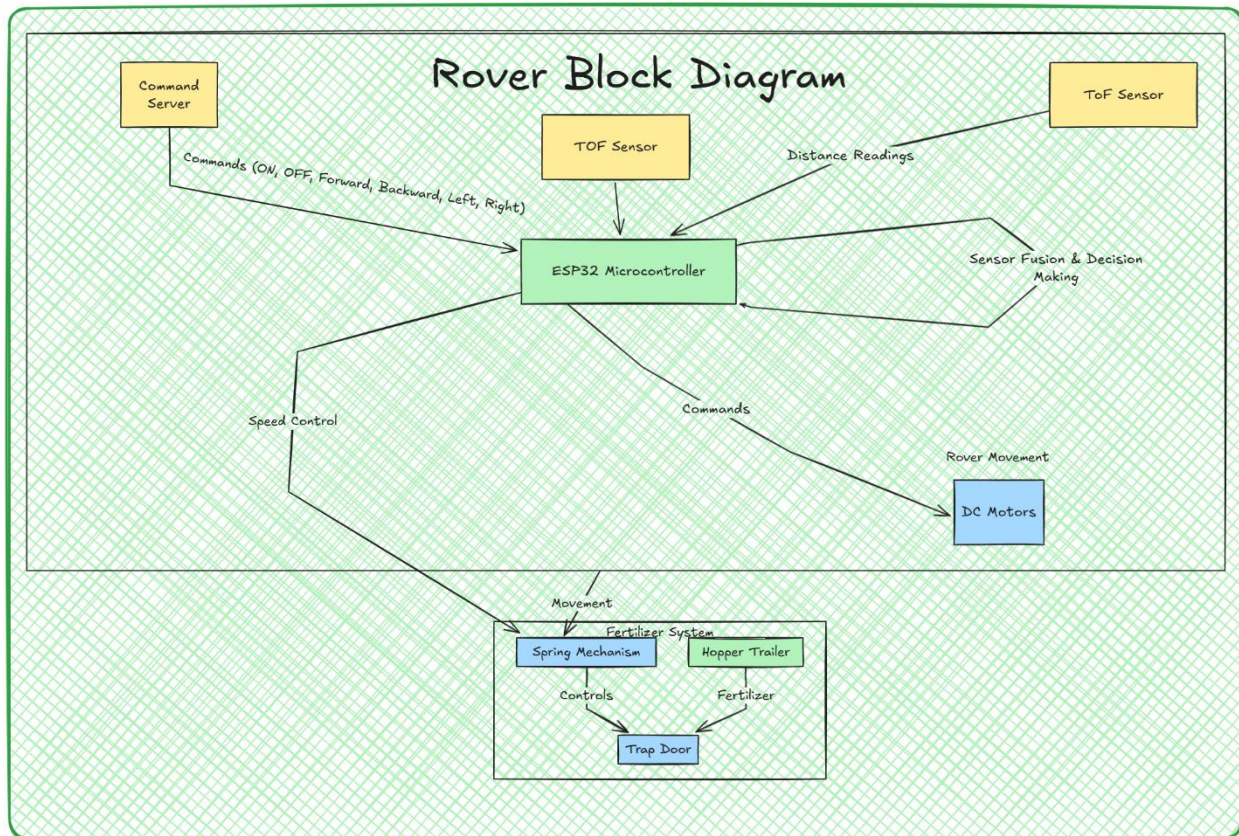
- b. Setup start/stop/left/right/backwards commands user can send to rover from command system
 - c. Setup TOF sensor as an emergency stop and autonomous left/right turning
- 4. Spread of material
 - a. 3D Model Hopper Trailer
 - b. Slice and print hopper trailer parts
 - c. Setup trap door for material spreading
 - d. Connect to rover
 - e. Iterate on design to allow for open/close that works consistently with rover pull action
 - f. Figure out stopping mechanism to provide friction to trailer so that trap door is opened.
- 5. Final rover design
 - a. Design modular rover chassis that fits all electronics and keeps them safe from the outside.
 - b. Design cycloidal gear system to turn the wheels from DC motor output
 - c. Design custom simplification for rocker-bogie (rocker no bogie)
 - d. Integrate springs and hitch into rover slide system
 - e. Integrate TOF sensor into obstacle detection for rover
- 6. Test full system integration

Detailed Task Breakdown

To start I plan on building out the motor control and object detection in tandem while I am waiting for a few final parts for the drive system. I plan on building up the object detection using an Ultrasonic sensor as the base which will be used to detect if the rover needs to stop and then a TOF sensor to determine whether the rover can try to navigate around the obstacle. I want to first test these systems by themselves and integrate them together with the ESP32 ESP-S. Once I have this to a working state, I plan on moving to communication between the boards using HTTPS while modeling the hopper trailer. I am going to build a basic communication interface that consists of basic commands dependent on object detection results and user commands from command app. I will do various system tests and integration tests to ensure everything is working individually and together. Afterwards, I plan on finishing up the design/building of the chassis and a better drive system for the wheels. For the drive system I plan on customizing a single tooth cycloidal gear box to be used with much larger wheels on the rover. Once I finish these parts I will test out the rover on its own and attach a tow hitch to connect to the hopper trailer. Then finally finish building the hopper and test out full system integration.

Description of Design Prototype

Overview

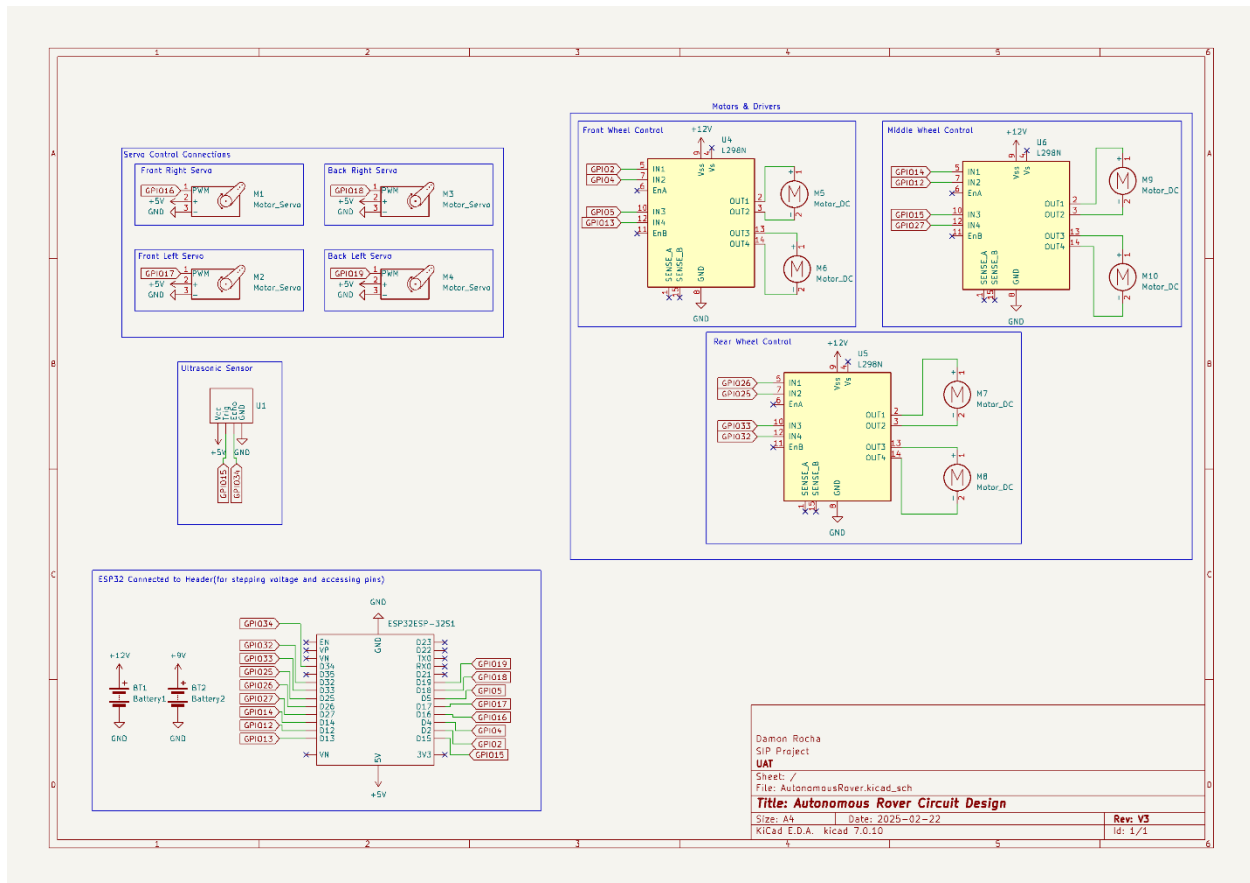


Appendix B

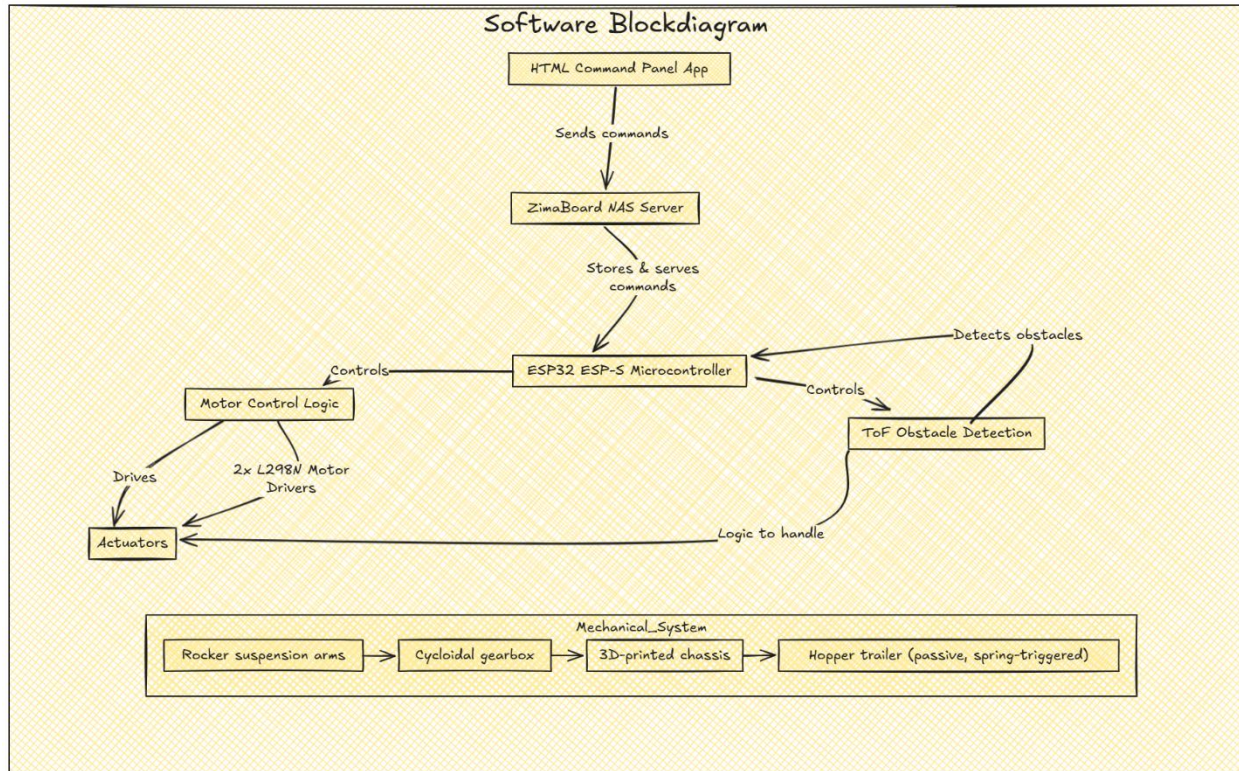
Electronics

I started the circuit design with the schematic below. However, I soon found that the 6V supply was not enough to get my rover moving on carpet let alone grass. I did not, however, have a 12V supply initially so I powered my initial prototype by wiring 2 6V packs in series and using a 9V battery to power the servos/microcontroller. I ended up scrapping 2 of the motors and the servos due to the circuit being overly complex and the added weight causing issues with pulling the hopper consistently. With my new gear system, I was able to use 4 motors to easily pull the trailer. I also used a power distribution board to help with overheating and stop any fires due to power direction switching because of the motor driver's H-Bridge. The last schematic is the final one for electronics.

The command queue server is Zimaboard NAS that hosts the server using a docker image. This allows the rover and camera to easily communicate through the NAS which was much more stable than the communications between the devices themselves.



Appendix C



Appendix E

Mechanical Design

The mechanical design of this system consists of the rocker suspension system, a set of 4 single tooth cycloidal gearboxes, the rover chassis, and the hopper trailer. This section will briefly cover the design of each of these items; starting with the rocker suspension system.

(IMAGES HERE OF OLD ROVER, V2 SUSPENSION, and FINAL SUSPENSION)

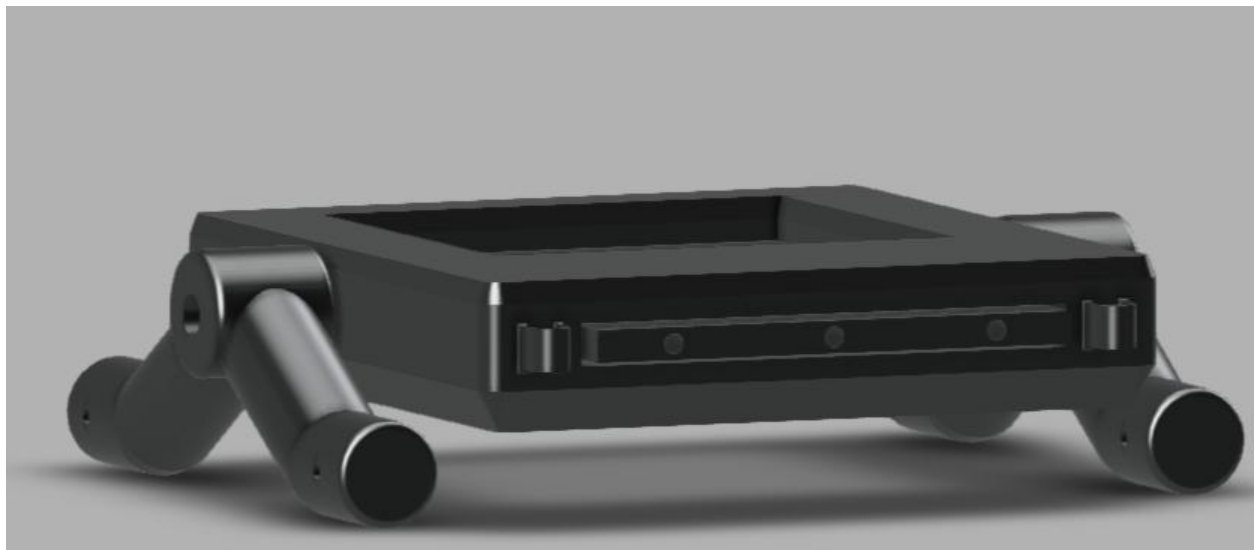
The rocker suspension system came from needing simplification of the more complex rocker bogie system. I had initially borrowed the 3d model of this from an instructables build and then after running into issues with pulling power and strength of the parts I designed my own. I found that this was too complex of a system and determined that I only needed the wheels on the left/right side of the rover to move independently to allow for traversal over uneven terrain. This led to the removal of the bogie and just using the rocker. Each wheel has its own motor so that each one can turn independently of the others. The arms that the wheels are attached to are mounted on a rod with a bearing to allow the arms to rotate forward and backwards, allowing the wheels to be on different elevations while keeping the same force from the motors. This allows for better pulling and traversal over a garden bed or farmland which can have uneven terrain. The next section covers the single tooth cycloidal gearbox and why it was chosen.

(IMAGES HERE OF OLD GEAR BOX, BROKEN ITEMS, GEARBOX ITERATIONS, AND FINAL CYCLOIDAL GEARBOX)

The first wheel drive system also came from the same instructables(INSTRUCTABLES) it was good to get my MVP done, test out my drive logic, and get some ideas on where to build my own. However, it was fairly limited for what I wanted it to accomplish. I broke quite a few of the motor housings and caused the wheel gears to strip anytime I had it pull extra weight. So I decided to look into compact gear systems that provide more torque. I found the Cycloidal gear system and from there a Single Tooth Cycloidal Gear system that was a great starting point to build my own gear box.

(CYCLOIDAL ITERATIONS HERE)

This system consists of a motor housing that mounts on a rocker arm. The motor housing is connected to a key that turns the single tooth cycloid. This cycloid then turns a modified gear that allows for the larger cycloid to roll across it as it turns. This gear is then connected to a wheel with bearings throughout the system to allow for smooth movement of the gears and wheels (RoTechnic, 2022). The wheels I initially used broke my system because they were much too small, so I upgraded to some bigger wheels that I did not design which worked perfectly for this project (AurthurGuy, 2018). Finally, for the last mechanical system the hopper trailer was designed from the ground up to distribute materials without any extra electrical components.



Appendix E

The Hopper Trailer is made up of a funnel attached to a slide system on wheels and was entirely built using 3d printed parts. The slide is held closed using elastic connected to a metal rod on the back of the hopper. To open it a hitch is attached to the slide where the rover can be connected so that when the rover is moving the slide is open and when the rover is stationary the slide shuts. An added benefit of the cycloidal gear box is it introduces a subtle natural rocking motion which helps distribute the material due to the slide slightly opening/closing as the rover is moving.

Evaluation Plan

This project will go through testing and evaluation across many different levels. Each system will be broken into modular parts and tested independently (unit tests) and in junction with other systems (integration tests). Each section of the evaluation criteria represents one system and will be tested as such. So, for example, the motor control of each wheel module will be tested independently and then together, then as a whole for the rover, then also integrated as part of the full autonomous system. On top of this the requirements I stated at the beginning of this brief either need to be met or show areas that need improvement. For clarity, here those are again:

- Drive on uneven surfaces
- Pull ~1lb of fertilizer
- Operate on 12V
- Evenly spread materials
- Autonomously avoid obstacles using camera and ToF
- Move ~ 2mph while not hauling spreader

Project Completion Assessment

Overview

In this project I set out to create a sustainable, autonomous rover inspired by mars rover designs geared towards accessible agricultural practices. Over the course of development, I successfully built and iterated on a modular rover platform capable of remote and autonomous navigation, equipped with object detection and a custom cycloidal gearbox/rocker drive system. Despite setbacks such as hardware failures, power issues, and weather-related delays, the project reached functional completion.

What went right

I had a lot success and learnings as part of this project some of the major items are the following:

- Single Tooth Cycloidal Drive System
 - The development of a custom single-tooth cycloidal gearbox significantly enhanced torque and introduced a natural rocking motion which improved soil

distribution capabilities. I did not plan on this at all it was just a bonus of the new system

- Effective Iterative Design
 - Transitioning from a complex rocker-bogie suspension to a simplified rocker system reduced weight and improved maneuverability. In this process I learned a lot about not being too rigid with plans and designs in the many iterations.
- Prototyping and 3d printing at scale
 - Utilizing 3D printing for components like the hopper trailer and camera mounts allowed for rapid prototyping and iterative improvements. Having a 3d printer and using it a lot was a bit of a waste of filament, but I learned a lot by modeling something then printing it out and seeing how it fared.
- Integrated Control Systems
 - I was really happy with the Zimaboard NAS docker command server and was able to integrate object detection using TinyML onto a microcontroller camera. While I did not keep the TinyML and the command server did start a fire when I first used it these were great learning experiences that lead to the success of the rover.
- Comprehensive Planning
 - I organized a lot more in tracking milestones, designs, and project tasks using a Gantt charts, block diagrams, and kanban boards. Normally I am all over the place with project tasks and this really kept me in check.

What went wrong

As for what went wrong, honestly there was a lot. I had to make a large pivot from heavy AI navigation to more traditional but reliable systems and many small pivots to get to where I am now. Some of the major items are as follows:

- Hardware Limitations
 - The Plastic gear TT motors I was using initially led to failures underload, necessitating a switch to more durable metal gear motors.
- Issues with original prototype rover models
 - The original rover models I got were great to get started on but were complex for what I was doing and broke quite easily. On top of that they were not made to be used with a load which probably contributed to the stripped plastic TT motors.
 - My second iteration tried to keep some of the parts of the original but would not work due to the wheels being too small and lead to me wasting quite a bit of material.
- Power Management Issues
 - When I first setup my server control system and app-based controller, I cause high current draw due to my code switching quickly between forward and backwards due to the commands which caused my battery wires to catch fire. Luckily, I was

able to put this out and did not lose my entire rover. I used a fused based battery management system from here that did not cause any more issues and allowed me to have a cleaner power setup.

- Environmental Challenges
 - Severe weather conditions in Texas caused power outages, took out my 3D printer and slightly affected my projects timeline. I ended up getting a much cheaper printer to finish my rover, but my original printer is currently a 3-axis shell.
- Connectivity Problems
 - My initial prototype of TinyML object detection and LLM based object detection worked but where unstable and way to slow for real time coms. I ended up having to pivot to accomplish object detection and avoidance which lead to the NAS based command server.

What would I change

If I could go back and change anything for this project I would have started way simpler and not used someone else's rover design. This gave me a bit of false confidence that led me to think way too big early on. I learned a lot in the process of trying different things out but if I had started with the essence of what this project is instead of trying to modify someone else's design to fit my innovation I think I would have had an easier time building this thing. However, if I had done that I would not have learned as much about TinyML, LLM based object detection, rocker-bogie suspension system, or cycloidal gear systems.

Future

If I take this rover further, I would like to make a bigger version of it using a raspberry pi 3. On top of that I would like to set up 360 degree lidar instead of a ToF sensor for better obstacle detection and overhaul the hopper trailer to use a more reliable force powered systems to open/close the slide. I really feel like a scaled version of this system could be used for many autonomous hauling tasks and after successfully showcasing material spreading I would try to look into other uses for this rover.

Appendices

Appendix A: Picture of Spirit Mars Rover – SpiritRover.jpeg

Appendix B: Rover Block diagram – RoverBlockDiagram.png

Appendix C: Circuit Design Schematic – SIPCircuitDesign.png

Appendix D: Circuit Design Schematic Final – SIPCircuitDesign5.png

Appendix E: Software Blockdiagram - SoftwareBlockdiagram.png

Appendix F: Hopper Trailer Final CAD model – HopperTrailerSide.png

Appendix G: References

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